2019-6-30 17:14

शोध संक्षेपिका

राष्ट्रीय संगोष्ठी

28 से 30 जनवरी, 2019

मंदिर स्थापत्य एवं शिल्प का विकास : 5वीं से 11वीं सदी ईस्वी तक

ABSTRACT

National Seminar

28 to 30 January, 2019

Development of Art and Temple Architecture: c. 5th to 11th A.D.

संकलन एवं संपादन सहयोग
राहुल कुमार सिंह
डॉ. प्रताप चंद पारख
प्रभात कुमार सिंह

आयोजक
संचालनालय संस्कृति एवं पुरातत्त्व, छत्तीसगढ़ शासन

2019-6-30 17:1

28	-	Newly Discovered Shaiva Establishment and Temples at Kundanpur, Tehsil Kolaras, District Shivpuri (M.P.)	:	Dr. Rajendra Yadav	
29	-	Brick structures from Eastern Vidarbha during 5th to 8th Century C.E.	:	K.S. Chandra	
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31	-	Architecture of Ambarnatha Temple in Kalyan of Maharashtra	:	Himanshu M. Mahajan	
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Brick structures from Eastern Vidarbha during 5th to 8th Century C.E.

K.S. Chandra, UGC NET, Contributory Lecturer
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Brick as a building material is being used traditionally, produced in numerous classes, types, materials, and sizes which vary with region and time period, and are produced in bulk. Two basic categories of bricks are fired and non-fired bricks. Fired bricks are one of the longest-lasting and strongest building materials, sometimes referred to as artificial stone, and have been used since circa 4000 BC. Air-dried bricks, known as mud bricks, have a history older than fired bricks, and have an additional ingredient straw as a mechanical binder. Bricks are laid in *courses* and numerous patterns known as *bonds*, collectively known as brickwork, and may be laid in various kinds of mortar to hold the bricks together to make a durable structure. (en.wikipedia.org). In the following paper we will view some of the Brick structures erected in Eastern Vidarbha during 5th to 8th Century C.E.

The earliest bricks were *dried brick*, meaning that they were formed from clay bearing earth or mud and dried (usually in the sun) until they were strong enough for use. The oldest discovered bricks, originally made from shaped mud come from the ancient Indus Valley cities of Mohenjo-Daro, Harappa, and Mehrgarh. Ceramic or *fired brick* was used as early as 3000 BC in early Indus Valley cities like Kalibangan. (en.wikipedia.org)

During the period 5th to 8th Century Vidarbha saw rise of a notable dynasty Vakataka which made its mark in Indian History through their Art, Sculpture and Architecture. Their Artistic tradition had affinity to Guptas from North. During this period Vidarbha saw some finest brick structures being erected in the form of fortification, dwellings and shrines. Some of them are Mandhal, Mansar, Nagara and Nagardhan in Eastern Vidarbha.

Chronological observation of excavated sites: Mandhal (20°57'11.64"N 79°27'31.69" E)

The Department of Ancient Indian History, Culture and Archaeology, Nagpur University, Nagpur, under Dr Ajay Mitra Shastri, assisted by Dr. A. P. Jamkhedkar, G. G. Deglurkar, O. P. Varma, and H. N. Singh undertook excavation at Mandal in Umred Taluka, in the season 1975-76 and 1976-77. Three brick structures, at Bhola Hudki (BHK-I), 1 km from the Mandhal, Bongi Hudki (BHK II) and Varhadi Talao (MDL-II) were exposed during the excavation. (*IAR* 1975-76:36; *IAR* 1976-77:39)

Mahabalipuram the Pallavas Architecture

Dr. Danteshwari S. Lakshme, Lecturer (English)
Govt. Higher Secondary School Pandripani, Jagdalpur, Bastar-Chhattisgarh

The Architectural contribution of Pallavas at Mahabalipuram dates back to 7th CAD. That Pallava kings Mahendra Verman and later Narasimha Verma II owe the credit for their marvelous contribution to the Indian Temple Architecture; which later laid the very genisis for the Dravidian style of Temple Architecture. We have classified our study among Pancha Pandava Ratha including Draupadi Ratha, The Shore temple, Descents of Ganges, Varaha caves, Tiger caves. As far as a Comparative Architectural study is concerned the art and Architectural features during the reign of Mahendra Verman are yet crude and rough and seems that of very preliminary stages while; Narasimha Verma II -Simhala are very fine. The Pallavas undoubtedly laid many marvelous features for the verynisis for the Dravidian Style of Temple Architecture.

Brick structures from Eastern Vidarbha during 5th to 8th Century C.E

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Abstract

Brick as a building material is, used traditionally, produced in numerous classes, types, materials, and sizes, which vary with region and time, and are, produced in bulk. Two basic categories of bricks are *fired* and *non-fired* bricks. Fired bricks are one of the long lasting and strongest building materials, sometimes referred to as artificial stone, and have been, used since circa 4000 BCE. Air-dried bricks, known as mud bricks, have a history older than fired bricks, and have an additional ingredient straw as a mechanical binder. Bricks are, laid in *courses* and numerous patterns known as *bonds*, collectively known as brickwork, and may be, laid in various kinds of mortar to hold the bricks together to make a durable structure. (en.wikipedia.org), in the following paper, we will view some of the Brick structures from Eastern Vidarbha during fifth to 8th Century C.E

Brief History

The earliest bricks were *dried brick*, meaning that they were formed from clay bearing earth or mud and dried (usually in the sun) until they were strong enough for use. The oldest Discovered bricks, originally made from shaped mud come from the ancient Indus Valley cities of Mohenjo-Daro, Harappa, and Mehrgarh. Ceramic or *fired brick* was, used as early as 3000 BCE in early Indus Valley cities like Kalibangan. (en.wikipedia.org)

Introduction

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Figure 1: Mandhal (Source Google Earth 2019)

Description of the structures:

Bhola Hudki (BHK-I) [Figure 2 A]

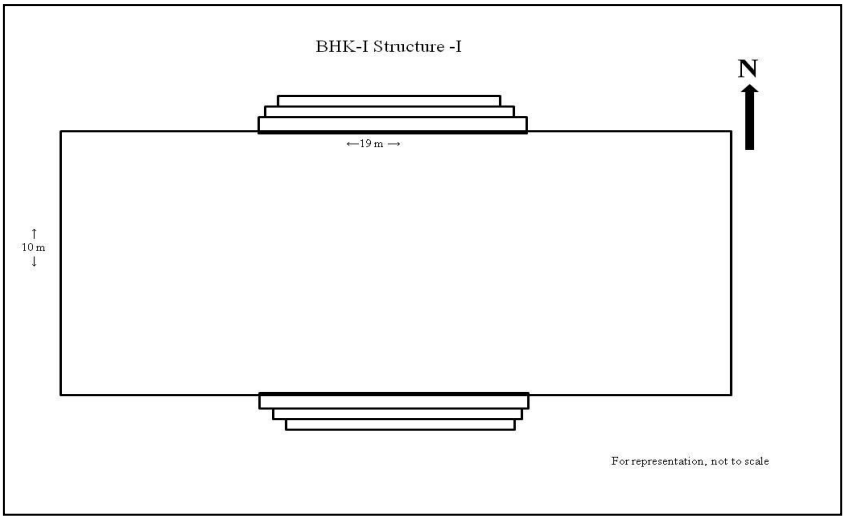
- Rectangular brick structure l9 x 10 m oriented east west with flight of steps on the north and south.
- Brick were of two sizes = 44 x 23 x 8 cm and 40 x 24 x 8 cm.
- Rectangular platform, measuring 18.0 x 10.60 x 1.20 m was exposed. In the middle of the platform traces of what appears to have been a small shrine was found, represented by an altar 2‘10 x 0.95 m with three courses of bricks, a semicircular brick platform to its west and a fragment of a wall going round the altar. A flight of steps on the north leading to the platform, a brick-built cistern 2.30 x 2.50 x 159 m on the east of the platform and a couple of brick chambers on the west were exposed.[**Figure 2 B]**

Bongi Hudki (BHK-II) [Figure 2 C]

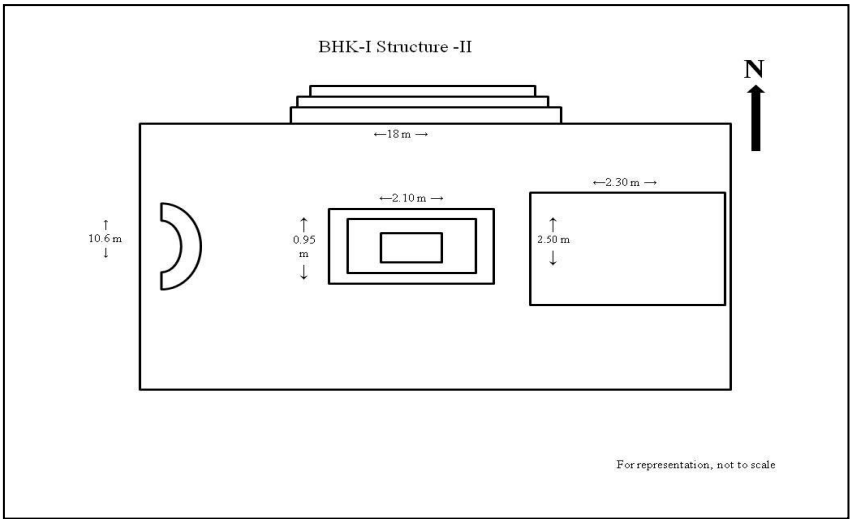
- Remains of another temple built on a massive platform 11.70 x 14.70 m had a *Garbhagriha* 4.50 x 5.60 m and *mukha-mandapa* 4.00 x 3.30 m.

Varhadi Talao (MDL-II)

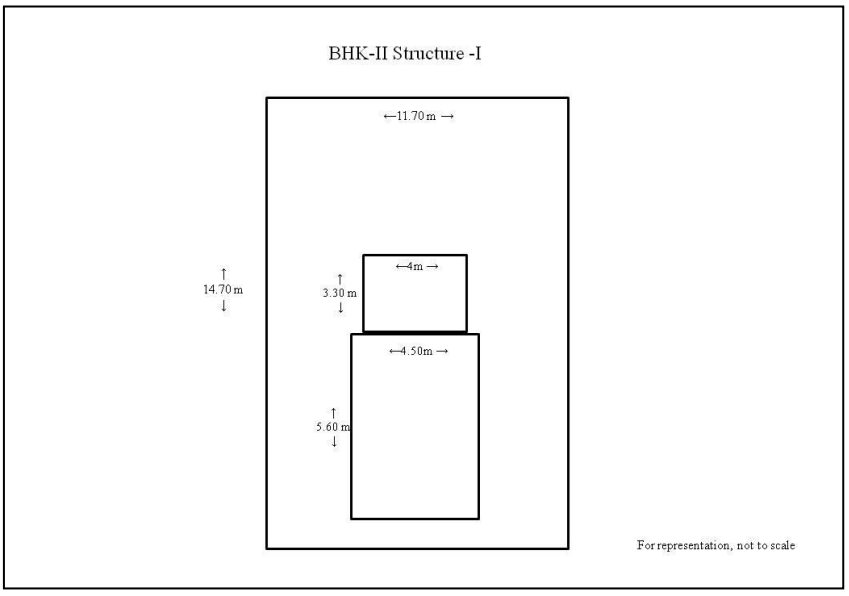
- Remains of a brick temple were uncovered originally built of massive brick walls, it was first reconstructed in the late Vakataka period or shortly thereafter. The square *Garbhagriha* had 8ft. 2in. thick walls preceded by a balustrade, 14ft. 9in. long flanking a flight of steps. The brick-built plinth was, encased in blocks of buff sandstone. In twelfth century C.E it was provided with rubble wall raised on the remains of the earlier stonewall
- A brick structure measuring 11.5 x 10.0 m, oriented north south, with a couple of projections on the west. At the bottom of the foundation of the structure, on the northern and southern sides, alignment of stones dressed on the exterior, were, noticed.



A



B



C

Figure 2: Representative Illustration of Mandhal Shrines – drawing by K.S.Chandra

Nagra (21º29’50.63”N 80º12’59.98” E)

The Department of Archaeology and Museums, Government of Maharashtra, under the direction of Dr A. P. Jamkhedkar assisted by Shri P. M. Muley, conducted excavation at Nagra, 6 km from Gondia on the way to Balaghat, in the season 1979-80 and unearthed remains of a brick-built temple datable to the Vakataka times. (*IAR* 1979-80:56)



Figure 3: Nagra (Source Google Earth 2019)

Description of the structure:

- The plinth, the only extant part of the temple having Length = 28.75 x 26.30 m, and Height =3 m.
- The outline of this rectangular plinth oriented east west is broken by Alternative projections and recesses that are respectively 2.15 and 1.45 m wide
- The vertical face of the plinth consists of different moldings at the base with niches on top.
- The niches on the projections rest on railing patterns and those in recesses have pilasters in their frame.

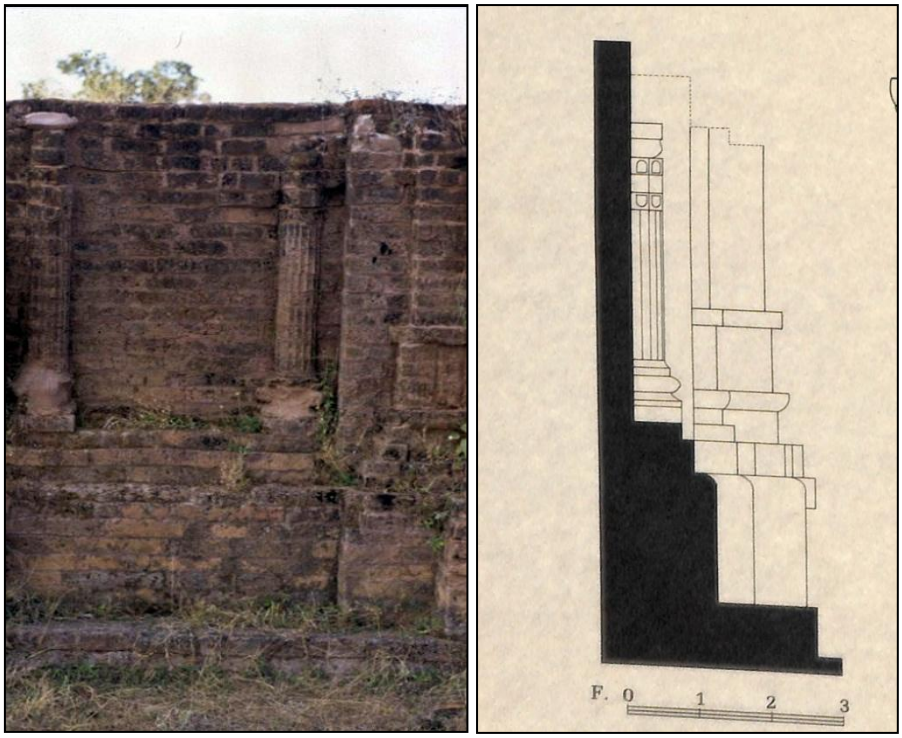
Another structure, a brick-built cistern with traces of a stone-cut channel spilling water to the cistern was, traced.



Figure 4: Rectangular plinth is, broken by alternative projections and recesses (front view) (Hans Bakker 2008:17)



Figure 5: Rectangular plinth is broken by alternative projections and recesses (side view) (Hans Bakker 2008:18)



A **B**

Figure 6: A-Nagra Brick temple moulded platform front view, (Hans Bakker 2008:20) **B-**Nagra Brick temple moulded platform sectional elevation (EITA Vol 2 Part 1 North India, fig.25:64)

Mansar (21º23'45.37"N 79º16'40.63" E)

Mansar represents one of the finest examples of Vakataka brick architecture in the form of palace, fortification and shrines. T.A. Wellsted carried out the first extensive survey of the site as early in 1933 (JPASB 1933: Vol XXIX), who brought to light antiquarian, Epigraphical and structural remains of the Vakataka period around *Hidimba tekadi*. To the east of this hillock, structural remains of a great monastery were, noticed. Initially The Excavation Branch I, Nagpur, of the Survey, under Amarendra Nath, assisted by N.C. Prakash, S. Prathapachandran, Ch. Babjirao, N.K.S. Sharma, P.M. Bhope, H.J. Barapatre, S.K. Gulrandhe, R.G. Nagurwar, D.K. Kasbi, D.S. Sambharkar, S.M. Khairkar, A.E. Keshwar, Shahid Akhtar and D.A. Kamble carried out excavation at Mansar in the season (*IAR* 1994-95:55-57). Then extensive Excavations were, carried out under the supervision of *Bodhisattva Nagarjun Smarak Sanstha va Anusandhan Kendra*, Nagpur, directed by Jagat Pati Joshi and A.K. Sharma, of Archaeological Survey of India assisted by Julfiquar Ali, Archana Dubey Asthana, P.M. Khobragade and J.S. Dubey in the seasons 1997-2000.

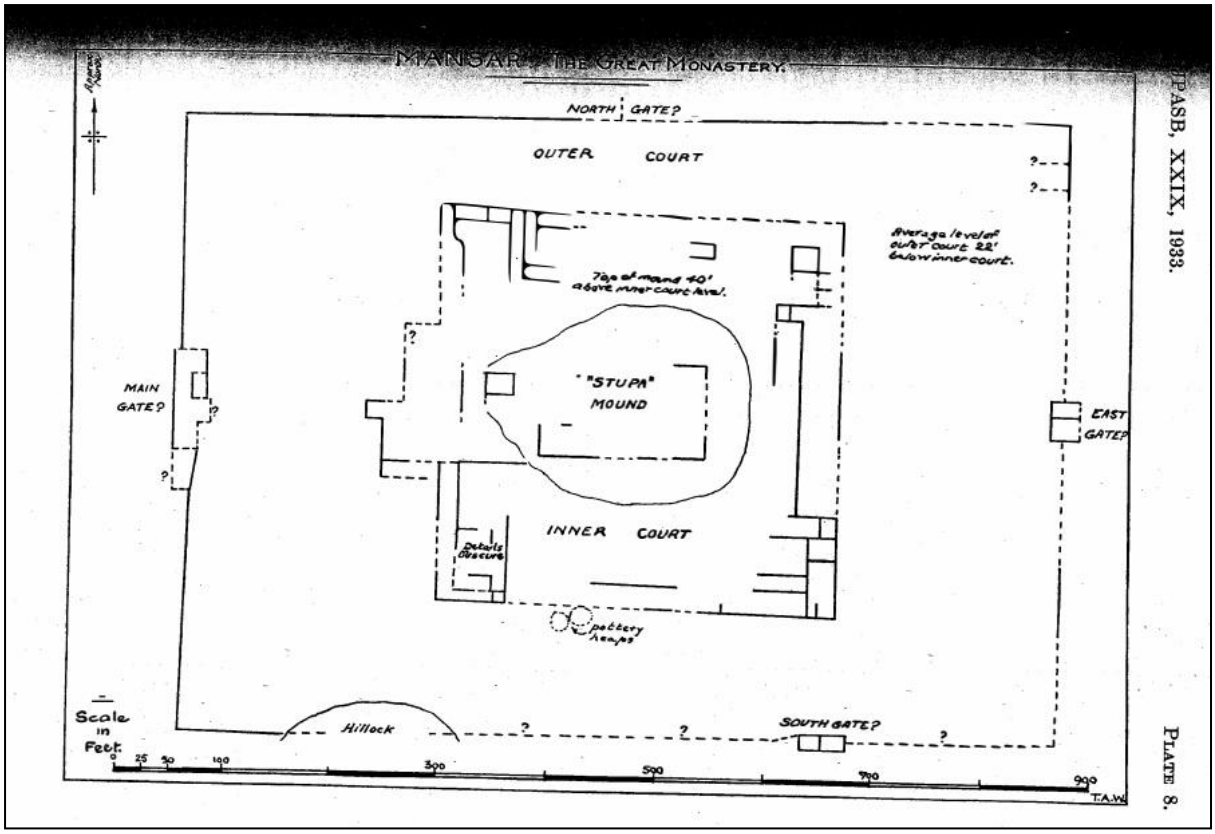


Figure 7: Wellsted's plan of Mansar (JPASB, XXIX, 1933: pl.8)

Description of the Structures:

Brick Shrines

During the initial phase of season, 1994-95 excavation revealed a portion of massive burnt-brick structural complex in mud-masonry with *ratha* projections, tentatively showing three phases of constructional activities.

The structural activities of **phase 1** were, noticed in the core area, showing western elevation of molded *adhishthana,* couple of *sopanas, pradakshinapatha* and an entrance to the complex. The core area seems to have been free-built over a knoll, taking advantage of its natural height, which in course of time proved disadvantageous to the structure as a whole on either side the flanking *sopanas* leading to *adhishthana* consisted of fourteen steps. A parallel wall, its pathway bearing traces of lime-mixed brick-jelly floor, demarcated the pradakshinapatha. The entire complex seems to have been, enclosed by a *prakara* with low balustrade entrance in the corner of western wall.

In **phase II** of construction, the *adhishthana* seems to have been, enlarged without disturbing the original ground plan of core area. The western exterior of the *adhishthana* is bedecked with offsets *(devakoshthaka)* and pilaster moldings.

In **phase III**, the *sopanas* leading to the floor of *adhishthana* were, found closed by raising regular walls across them and ultimately filled with brickbats. Above the *adhishthana,* portions of a porch and subsidiary shrines on its northern and eastern segments have been, noticed. The floor of the porch had thick-burnt deposit suggesting one of the possible reasons of decay of the establishment; two *in situ* remains of burnt wooden-post bear testimony to a large-scale burning at the site. The shrine to the north consisted of three multi-faceted pedestal-like platforms while the shrine of eastern segment comprised dilapidated pedestals with sealed clay images largely damaged. A few large-sized standing clay images of divinities and *dvarapalas* were, found in shambles with crest-fallen. A few sculptural fragments, which could survive the onslaught, were of great iconographic value in determining the religious affinity of the structure largely. In order to protect the massive free-built core area of the complex from the vagary of nature, it appears that oblong box-like construction of ramp-walls was, envisaged on the western flank. Each one of the ramp-walls emerging at regular intervals, at right angle, was an offshoot of a common wall of shorter axis, ultimately terminating against the western profile of *adhishthana* in support of core area. It seems a natural gradient was, given from core area to further west towards the shorter axis wall. Subsequently, the space between the walls appears to have been, systematically filled with stone chips and ferruginous clay, again maintaining the natural gradient. Unlike the entrance of **phase I**, a massive burnt-brick pillared entrance, asymmetrical to the enclosed shrine, was, noticed in the cuttings. It raises the possibility of tumbling upon some other structural complex stratigraphically anterior to the one noted above.

During the extensive excavations in season 1997-2000, the following structures were, exposed completely: Two sacrificial altars, one in the shape of a *Syenaciti* and the other a *Kurma chiti,* made of bricks. The entrance to the *Syenaciti* and *kurmachiti* was exposed. It is 2.90 m in width with three flights of steps. The lower two lying in the center are 0.80 m in width. The entrance has two side arms. While exposing the entrance, a brick wall was, noticed, raised later that had collapsed all at a time. The *Syenaciti,* which is in the north, was made after smashing the cross walls of the boxes of the earlier *stupa,* which has gone out of use by that time making the ground, leveled. In the *Syenaciti* figure of a *Purusha* made of lime was, found sacrificed with his head smashed. A *vedi* in the chest portion with a hole for fixing *ayashti* over it was, made and a lamp was, found kept nearby. Two pots were, kept near the knee region of the figure on his right side with an iron snake kept near his left toe looking towards the human figure.



Figure 8: Mansar (Source Google Earth 2019)



Figure 9: Mansar *Syenaciti* figure of a *Purusha* made of lime (Hans Bakker 2008)

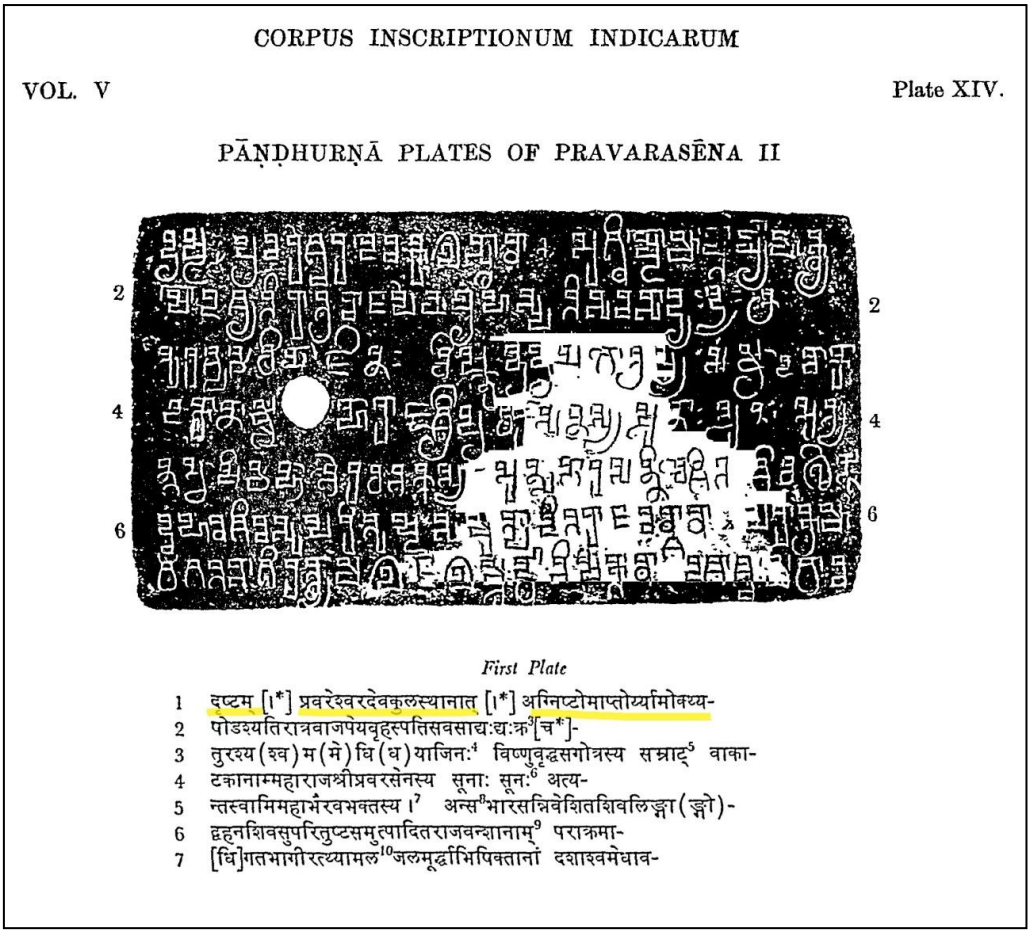
Purusa-medha was a simple stone temple with a *mandapa* and a *Garbhagriha* on the western side of the mound. Adjoining the *kurmachiti* in the southern side there is a stone staircase comprising eight steps leading towards a small *Havana kunda*, a limekiln and a square shrine made of bricks, which apparently appears to have been a *Saiva* shrine as there is a hole for outlet of water. Stratigraphically, it appears that the sacrifice was, made at the end of the Satavahana period and the beginning of the Vakataka period. A whitish layer seals the sacrificial altars and other associated structures. During this period, on the eastern side of the mound, a magnificent temple with a, plinth of dressed sandstone blocks and superstructure of bricks was built on the top by the Vakataka. This temple with two phases of construction has yielded many fragmentary sculptures having impressive headdresses and bedecked with jewellery. These are in the best traditions of the Vakataka art. Besides, a coin of Western Kshatrapa king showing Swami Rudrasena III's bust on the obverse and on the reverse a *Chaitya* with inscriptions reading '*Maha Kshatrapasa Swami Rudradaman Putasa Raja Maha Kshatrapa Swami Rudrasena* 21 Saka' (348-378 C.E) was discovered. Sculptured and architectural fragments on white stone, rough grained sandstone and fine-grained rose coloured sandstone were, recovered from dig during this period. (*IAR* 1997-98:129-133)



Figure 10: Mansar view of the complex **(**By Savearth - Own work, CC0, https://commons.wikimedia.org/w/index.php?curid=11980766)

In the season 1998-99, a sizable temple-complex was exposed. It is quite evident that the Siva shrines exposed belong to two phases. Taking advantage of the massive retaining walls built of dressed and undressed stone blocks and bricks measuring 46 x 27 x 7 cm were, raised to make a flat platform on the top of the hill for the construction of a temple and later a stupa. Around the temple in the first phase, the shrines were, mostly constructed on the northern side where two natural caves exist. One of the caves was also converted into a shrine by laying bricks on the floor and plastering it with lime-plaster and then applying over it red-ochre paint, while the other cave was used as a meditating chamber by constructing concave brick walled approach, the wall ultimately closing on the top. The shrines were, built over fluted base with sharp angles making octagons. Over these octagons, simple rectangular shrines were, raised, each having a niche facing north for the deity and having an approach from either the east or the west. Right from the bottom of the octagonal base, a straight staircase led up to the outer side of the shrines and later on to the cave shrine. The foot of this staircase had a brick-built semicircular *Chandrasila,* whereas two narrow staircases lead to the cave shrine apparently to maintain one-way flow of devotees. In order to close the roof of these staircases, triangles and inverted triangles have been, used in the construction. The walls and the steps were near the steps. In one of the shrines, the *linga* has been, kept over a square brick pedestal just below the natural rock inclined. On the southeastern corner at the base of the steps, two square *hawan kundas* (Fire Altars) were exposed. A large number of fragmentary sculptures and architectural members of temples mostly of red sand stone have been, recovered from the debris sealing these shrines.

The most noteworthy find clay sealing on the northeastern corner, just at the base of the northern row of shrines bearing the legend in box-headed Brahmi character, reading *Pravareswarasya* referring obviously to *Pravaresware deva Kulsthanat* referred to in the well-known copper plate inscription from Pandhurna. The main shrine was obviously the *Devakula* temple, the dynastic temple of the Vakatakas made by Pravarasena II (420-450 C.E) in memory of his great grandfather Pravaresvara I. Incidentally, the Pandhurna Copper Plate mentions its issuance from *Pravaresvara Devakulsthanat* or Pravaresvara Temple (V.V.Mirashi, 1963:65, pl.XIV) [**Figure 11]** which has now been unearthed at Mansar (ancient Pravarapura). The additions in second phase were, probably carried out by Prithvishena II, who was a devout Saivite. Apart from a number of sealings that bear the legend *Pravareswarasya* and *Shri Jaivirdhi,* some sealing depict elephant, the animal that appears to have of great importance to the Eastern Vakatakas.



CORPUS INSCRIPTIONUM INDICARUM

VOL. V Plate XIV.

PĀṆḌHURṆĀ PLATES OF PRAVARASĒNA II

First Plate

1 दृष्टम् [।*] प्रवरेश्वरदेवकुलस्यानात् [।*] अग्निष्टोमाप्तोर्य्यामोक्थ्य-
2 षोडश्यतिरात्रवाजपेयवृहस्पतिसवसाद्यःक्रः[च*]-
3 तुरश्य(श्व)म(मे)धि(ध)याजिनः[4] विष्णुवृद्धसगोत्रस्य सम्राट्[5] वाका-
4 टकानाम्महाराजश्रीप्रवरसेनस्य सूनाः सूनः[6] अत्य-
5 न्तस्वामिमहाभैरवभक्तस्य ।[7] अन्स[8]भारसन्निवेशितशिवलिङ्गा(ङ्गो)-
6 द्वहनशिवसुपरितुष्टसमुत्पादितराजवन्शानाम्[9] पराक्रमा-
7 [धि]गतभागीरत्थ्यामल[10]जलमूर्द्धाभिषिक्तानां दशाश्वमेधाव-

Figure 11: Pandhurna Plate of Pravarasena II (V.V.Mirashi, 1963:65, pl.XIV)



Figure 12: Mansar top view of rampart (By Savearth - Own work, CC0, https://commons.wikimedia.org/w/index.php?curid=11980766)

The lotus bud pattern of shrines discovered at Mansar display superb example of *vastukala* and construction of shrines without any regular, flat or domical roof as the ends of top petals joined and made the *sikharas* of the shrines well plastered with lime-plaster, also using lime concrete for strengthening the surface of the steps. Red sandstone sculptured pieces mostly belonging to Saiva cult has been, recovered from these shrines. On the western and southern sides a row of Siva, shrines built of bricks were exposed. These shrines built in three terraces are, approached by flight of steps. Out of total sixteen shrines exposed, six had *Siva-linga,* one each resting over a square brick pedestal having a *pranala* on the northern side. Whereas in some only pedestals are present. The pedestals have been, made out of chiseled-bricks. One of the shrines was, built in the form of a *yoni pitha* itself with the *linga* in the centre of the shrine over a brick pedestal. Vertically all the shrines were, built in the form of lotus buds by adopting the triangles and inverted triangles (equilateral). Method for raising petals that ultimately close into a pointed end making the closed roof of the *sikhara* was, adopted. The whole complex was, plastered with lime-mortar. At the base of these shrines, on the southern corner of one of the steps, a burnt wooden *yupa* was, found. The triangles, particularly on the outer side *of the jangha* portions of the row of shrines, make a human face. Each shrine has an approach from either east or west, with some having a small *mandapa* and small semicircular or triangular *kunda* for water storage (*IAR* 1998-99:114-116).



Figure 13: Mansar Square *Hawan Kund* with circular brick lining (Hans Bakker 2008)

In the final season of the excavation, continuation of the earlier work revealed the dressed-stone foundation of the *mandapa* and *Garbhagriha* of the earliest temple, facing east which was, partly exposed in 1997-98. It has a rectangular brick platform in the western wall of the *Garbhagriha.* The side entrance in the north is 0.90 m in width and comprises five courses of dressed rough grained whitish sandstone with two steps. North-south width of the temple is 12.60 m. In order to strengthen the northern and southern slopes of the hillock, massive pitching walls were, raised to an extent of 60 m, on each side using huge boulders. By the side of huge natural *granitic* outcrop was dug a cave on the southern side with its sidewalls lined with bricks, at a depth of 1.5 m. This has corbelled-arch to cover the top. Steps were, laid in bricks on the western side of the cave. To the west of this cave, two square *hawan kundas,* with brick linings were, exposed in front of the Siva shrines. The larger *hawan kunda* contains two rows of brick linings, outer 3.0 x 3.00 m while the inner one 1.35 x 1.35 m, the outer lining have two *pranalikas* on each side having a width of 0.20 m. Thickness of the lining walls is 0.85 m. Further, on the western side, on the foot of the Siva shrines, a huge hall measuring 9.60 x 9.60 m with attached verandah, 8.0 x 1.60 m and a corridor on the southern side leading to the main entrance of the cave shrines was exposed. While constructing these structures of bricks, the huge granitic outcrop on the northeastern side was not disturbed. The southern and eastern walls comprised two projections and decorated with moldings in the entire length. All the bricks have fallen in brick-on-edge position with mortar in-between remaining intact.

Apart from the structural remains, the most outstanding finds of season's dig were the most beautifully carved red fine-grained sandstone figures comprising Saivite images, particularly belonging to Tantric cult. The well-delineated sculptures of Siva-Parvati, Uma-Mahesvara, *ganas,* Lajjagauri and structural fragments are noteworthy for superb artisanship of Vakataka artists. A good number of inscriptions on structural fragments in typical eastern Vakataka box-headed Brahmi characters were, recovered. An earlier inscription datable to second century BCE was, recovered. The inscriptions are mostly prayer hymns in Sanskrit. A number of iron objects including dowels, nails, doorjambs etc., glass pendants, beads and other terracotta objects were, recovered. The structures exposed during the last two seasons were, duly conserved using the recovered bricks from the site [**Figure 15**]. (*IAR* 1999-2000:103-106),

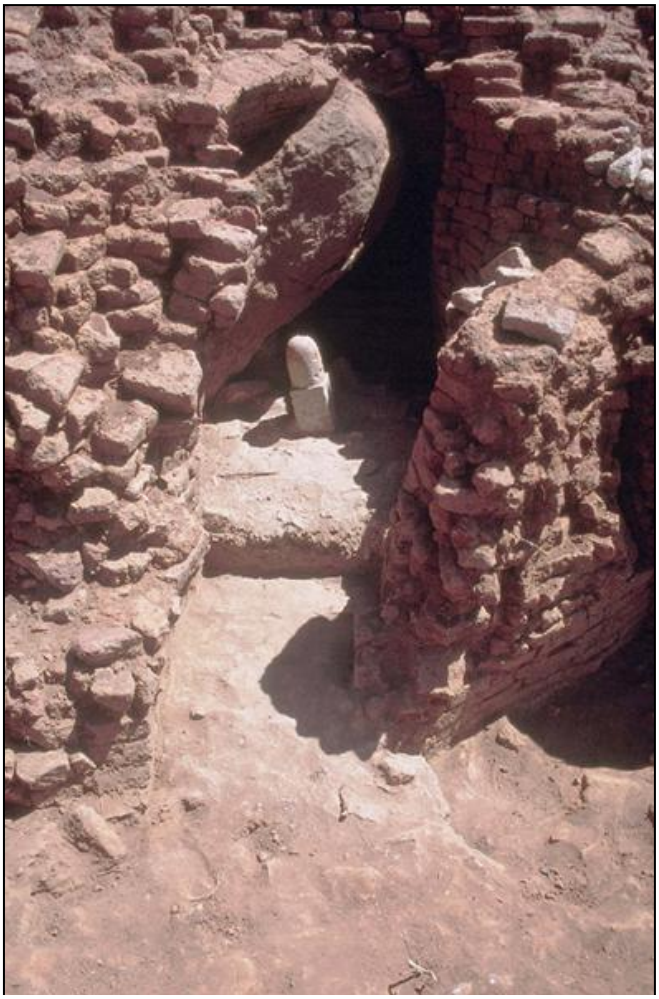


Figure 14: Mansar Siva Linga (Hans Bakker 2008)



Figure 15: Mansar Northwest Adhishthana (Hans Bakker, MNS2 2006: Fig 34)

Stone and Brick Stupa

After the Vakataka activities ceased, particularly keeping in view the commanding position of the hill, the Buddhists finally built a stupa having twenty-four brick-built boxes with a central box, all filled with rubble and stone. It is having a diameter of 18 m and steps on the eastern side. This large stupa was, built on the top and remained in existence until the end of the seventh century C.E. This stupa was also associated with an oval *Chaitya* on the southeastern side, which was, built over an earlier *Chaitya.* Again, on the eastern side a large impressive staircase has been, exposed for approaching structures built in different periods with its length and height increased through the centuries. The staircase also shows the evidence of the use of temple material in the stage of its renovation and enlargement during the time of the last stupa. Stone stupa built of semi-dressed stones had a diameter of 13.0 m and enclosed by a 1.10 m wide *pradakshinapatha* of bricks measuring 42 x 22 x 7 cm in size. The inner side of the stupa was strengthened by providing two stonewalls, running in east-west direction. On the eastern side, stupa height is available to a height of 2.10 m with six courses of semi-dressed stones. To the east of the grand steps and amphitheater, on a leveled surface, a *Vihara* made of bricks was, exposed up to twelve courses. It comprises at least nine rooms measuring 2.20 x 2.20 m each, on the north, south and east, made by partitioning an earlier 12.50 x 12.50 m hall. In-between the two rows of rooms a 8.00 m long east-west and 2.30 m wide, north-south, open space leads to a 7.00 m long east-west, and 2.10 m wide, north south, prayer hall, Bricks, used in the construction of this *Vihara* are 30 x 22 x 7 cm in size. While exposing the boulder foundation of the uppermost stupa, from the southeastern corner, at a depth of 1.10 m below surface, fragment of an earthen relic casket along with two fragments of charred human bones were, recovered. The relic casket of micaceous clay bears an additional coating of mica on the outer surface (*IAR* 1999-2000:103-106).

Mansar gives the succession of culture from *circa* 200 BCE to CE 700 (*IAR* 1997-98:129-133).

Nagardhan (21º20’14”N 79º18’59”E)

Nagardhan is a village in Ramtek taluka of Nagpur district is located 6 km south of Ramtek, the Taluka headquarter. This place has been associated with *Nandivardhana*, the capital of one of branches of the Vakataka dynasty that ruled between 250-550 CE. Earlier (Russel R.V, 1908:307-8), (Hiralal 1908) and later (Mirashi 1957) identified present-day Nagardhan village as historical *Nandivardhana* due to structural remains of the Vakataka period, and finding of inscription of Prabhavatigupta (daughter of the Gupta emperor Chandragupta II) in the Keval Narsimha temple at Ramtek. On background of archaeological and historical information the Nagpur division of the Department of Archaeology, Maharashtra State, and Department of A.I.H.C. and Archaeology, Deccan College, Pune jointly excavated the site of Nagardhan in two seasons of 2015-2016 and 2016-17.

The Brick structures were, encountered in Trench H (Quadrant1), Locality 2, 3, 4, 6



Figure 16: Nagardhan (Source Google Maps 2019)

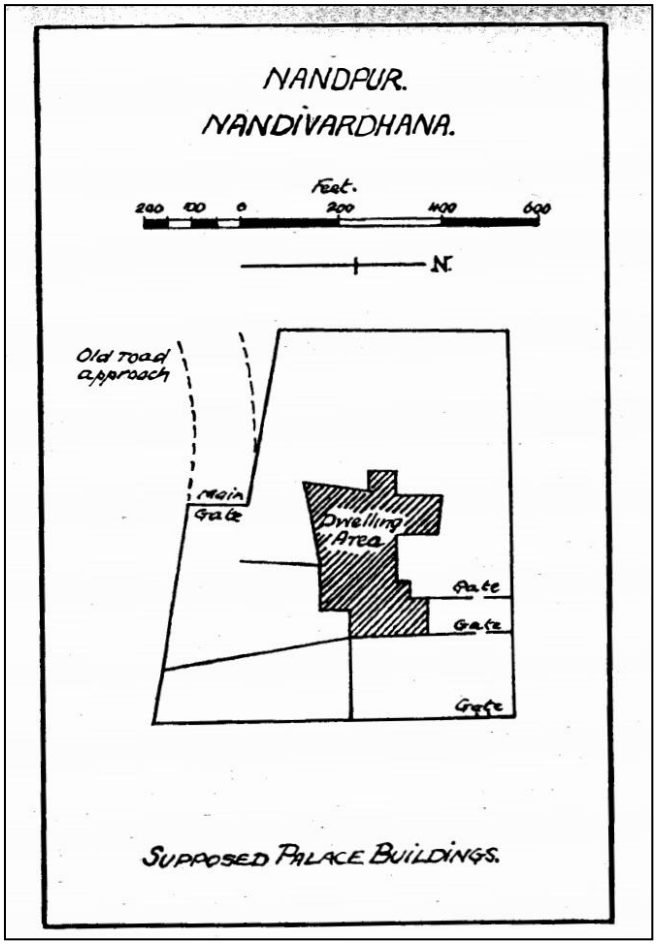


Figure 17: Nandivardhana (Wellsted JPASB, XXIX, 1933: pl.6, fig.2)

Description of the Structures:

Trench H (Quadrant 1) was, laid inside the fort to trace the cultural sequence. It provided evidence of several structural phases. A large structure of eight brick courses was found which, happens to be the earliest structural phase (Satavahana-Kshatrapa period) inside the fort. Below this level, earlier habitation deposit was, found. Ceramics of the 3rd-4th century BCE were, found just below the brick structure. The earliest habitation deposit showed presence of ceramics of the Early Iron Age affinity. Above the Satavahana-Kshatrapa, structural level a large brick structure was, found. It is eight extant courses over a bed of pebbles. This structure perhaps had a large room with an adjoining smaller room. Clay sealing bearing the name of Vakataka Queen Prabhavati Gupta was, found over this structure. Other clay sealing having the *Srivatsa* symbol was, found in the late Vakataka period deposit in this trench. A brick well was, found in the southeastern corner of the trench in this phase [**Figure 18 and Figure 19**]. The upper eight courses are made of bricks in header pattern.

The well wall up to 1.5 m depth is made of bricks. The shape of the well up this point is circular. The well section becomes square due to a single stone course. The wall below this stone course is somewhat disturbed. The wall construction using bricks continues down for 2.5 m. above the depth of 45-50 cm below datum point stratified deposit of the medieval period was, encountered. A few structures of reused bricks and stones were present. The bricks were half-cut portions of Vakataka period bricks.

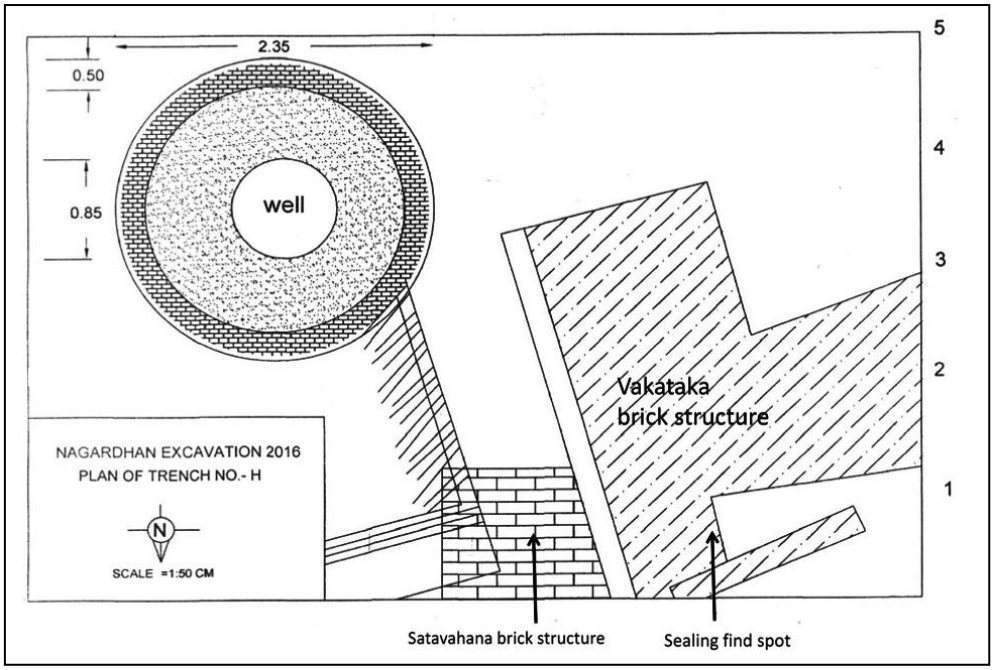


Figure 18: Brick Structures drawing (Sontakke et.al Fig.3)



Figure 19: Brick well in SE corner of the trench (Sontakke *et al.*, Pl. 4**)**

Locality 2: The trenches I, I1 and J in the Locality 2 (south of fort) also yielded good structural evidence. The lowest deposit belonged to the Early Iron Age level above which remains of the Mauryan period were, recorded. Both these phases had simple mud floors. Above the Mauryan period deposit, a structure resembling a pavement was, encountered. This pavement had only a single course of bricks laid over a rammed floor made by brick powder and lime and lateritic granules. Based on ceramic evidence it is clear that this pavement belonged to the Satavahana-Kshatrapa period. A Vakataka period brick structure of eight extant courses was, exposed that was, destroyed to some extent due to upper structural activities. This was only a single wall with a lime-plastered floor. This structure seems to be of public utility structure such as a courtyard. Several beads (stone and glass) and glass bangles were, recovered from this structural phase. The uppermost level dated to around seventh to 10th century C.E yielded evidence of another pavement. This pavement has 'Z' shape [**Figure 20]**. The areas besides the pavement were, filled with *murrum*.



Figure 20: 'Z' Shaped pavement (Sontakke *et al.*, Pl. 5)

Locality 3: Structures of the Vakataka and Late Satavahana phase were, found in trenches L and M in Locality 3 [**Figure 21**]. The first and earliest phase had small but well-made brick structures and only one long wall with perpendicular smaller walls, and a drain were, found. These structural remains were domestic in nature since antiquities like religious images, coins, bangles, beads and were, recovered here. In the second phase, one very long wall of 13 m and of six brick courses was, found. A well-rammed floor made of crumbled bricks, lime and lateritic granules was associated with this phase structure. Seven walls, nine alignments, one pavement and two partition walls were, found in the upper phase.



Figure 21: Brick Structures in locality 3 (Sontakke *et al.*, Pl. 6)

Locality 4: From here, two full pots [**Figure 22**] and three full pans along with several artifacts were, found. These include a fragmented votive shrine, votive tank, iron chisel, a stone tile depicting a deer, and terracotta bangles. In addition, antlers of the spotted deer were, found here. The pots were, placed by deliberate cutting of thick layer of *murrum*. A rectangular brick structure was, constructed around one of these pots specifically for protection. Artifacts mentioned above were found in a hole (1.5 m deep) below the pots. The hole was, cut in the *murrum*. The arrangement of the pots, the manner of placing these pots and the material found there suggests that these were, connected with some ritual of the Vakataka period.



Figure 22: Probable ritual place during Vakataka time (Sontakke *et al.*, Pl.7)

Locality 6 This locality is embankment area. Four step trenches were, laid here. Excavation revealed interesting clues to how the embankment was constructed [**Figure 23**]. The lowermost deposit was of *murrum* followed by clay. This clay contained remains of the Early Iron Age habitation. This clay was, probably brought and deposited here perhaps during the Vakataka period. Outer side of the embankment showed presence of brick (two courses) alignment. Above this, *murrum* followed by clay and again brickbats were dumped.



Figure 23: Brick Embankment (Sontakke *et al.*, Pl. 8)

Second Season 2016-17

A 16-metre-long wall was, exposed, which was part of a palatial structure; the space is equivalent to the size of around five houses in the area. A lime-plastered platform with brick enclosure indicates a site of gathering or a market place, as glass beads and semi precious stones such as carnelian and agate were, found.

Discussion

The Extensive usage of Bricks by Vakatakas in construction was due to ease of handling, molding, light weight and uniformity especially in manufacturing, raw material- clay, sand and fine silt sourced from nearby ponds and hillocks. Construction with bricks was less strenuous compared to stone structures, which took heavy toll on work force. Brick laying was much easier and the areas where we find brick structures have sturdy hard surface and rocky underlay and it also helped them to bear the harsh weather of the central plains. This might have been one of the reasons for using large quantities of bricks in their earlier construction, which started a slow decline during their later period.

In the brick structures from Mandhal to Nagardhan, there is a continuous change in structural artistry; the brick structures described from Mandhal **[Figure 2]** seem to have been, constructed in a very articulate way portraying the grandeur of Vakataka architecture. The same is, seen in the plinth remains of Nagra brick temple, which gives us a fair idea about the entire structure when it was standing erect. The tradition continues in the form of massive fortification and Saivite temple complex along with *Syenaciti* and *kurmachiti* at Mansar identified as the capital of Pravarasena II (V.V.Mirashi, 1963:65, pl.XIV), which was bustling with activity during his reign. At the same time commercial and socio cultural activity was, seen in Nagardhan 10 km away where, brick structures were erected in the form of ritual places, wells, embankments, recreational area. The structural change observed here is that there is no much of artistic element, as seen at Mandhal, Nagra and Mansar. At Nagardhan, brick structures were in straight line, very few intricate design forms are, found here. The reason might be that Nagardhan must have been a place for social and commercial activity and for residence of ordinary people during Vakataka period. However, it does show a decline in architectural activity in the design and decoration aspect; which may be due to continuous occupation of Nagardhan after decline of Vakatakas, as excavations have revealed seven periods of cultures. Further excavations at Nagardhan and its surrounding area may give us more insight into this in future.

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Abbreviations

EITA – Encyclopaedia of Indian Temple Architecture

IAR – *Indian Archaeology –A Review*

JPASB - Journal of the Asiatic Society of Bengal

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